AEGIS: No Tool Call Left Unchecked — A Pre-Execution Firewall and Audit Layer for AI Agents

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Abstract

Static per-call benchmarks are how the field measures defences against malicious agent tool calls. We audit that practice and find it does not measure what it claims. The two corpora these benchmarks are built from—InjecAgent and ToolEmu—are agent-execution suites: neither ships a concrete attacker tool call, because that call only exists once an agent has interpreted an injection. Converting them to static per-call data therefore requires fabricating the call, and we show the fabrications in use are degenerate. In a composite benchmark of 5,525 records, 3,340 carry no tool call at all, a further 2,093 are exact duplicates, and the surviving malicious records encode their own label in an argument key: a single feature, argument character count, separates the classes at AUC 0.9944. Effective sample size is 92 unique evaluable calls, not 5,525. We rebuild the benchmark the only way that is sound—by running an agent. Replaying each InjecAgent case and recording what the model actually emits yields 663 records whose arguments are model-chosen, with no single feature above AUC 0.585 and every attacked tool also appearing in the benign class. A matched control arm that swaps the injection for length-matched neutral prose isolates the causal effect: attack success 95.1% versus 0.3%, a risk ratio of 334. On this corrected data the standard defence collapses. Classic payload patterns—SQL injection, path traversal, shell injection, sensitive files, prompt injection—match 0.4% of real attack calls, because a successful indirect injection produces a syntactically innocent call such as ViewSavedPaymentMethods({}). We argue payload inspection is the wrong abstraction for agent security: what makes such a call an attack is provenance and authorisation, neither of which is in the call string. Benchmark, rollout harness, and audit are released under MIT license.[1]

1 Introduction

LLM agents that act through tools (Yao et al., 2023; Schick et al., 2023) are moving into production, interacting with databases, file systems, and cloud infrastructure via frameworks such as LangChain (Chase, 2023) and CrewAI (CrewAI, 2024). A single hallucinated or prompt-injected tool call (Greshake et al., 2023) can cause irreversible damage before any human is aware. Yet in most agent stacks, no enforcement layer sits between the model's tool-call decision and execution.

How the field measures defences. Post-execution observability platforms (Langfuse, 2024; Arize AI, 2024) record events after side effects occur. To evaluate a defence that acts before execution, the standard move is to assemble a static corpus of labelled tool calls and report block rate and false-positive rate on it. The calls are drawn from public agent-safety suites, principally InjecAgent (Zhan et al., 2024) and ToolEmu (Ruan et al., 2024). This paper is about what happens when that assembly step is examined.

Neither corpus contains the attack. InjecAgent and ToolEmu are agent-execution benchmarks. InjecAgent ships the instruction an attacker hides in a tool response; ToolEmu ships a red-team case description. In both, the malicious tool call exists only after an agent reads the injection and decides what to do. A static per-call corpus therefore cannot be extracted; it has to be fabricated, and the fabrication is where the measurement breaks.

We audit a composite benchmark built this way (§3). Of 5,525 records, 3,340 carry no

[1] Code: https://github.com/Justin0504/Aegis.

tool call at all and are unscoreable by construction; 2,093 more are exact duplicates. The malicious records that remain were synthesised by placing the injected sentence under an argument key named _attacker_instruction, present in 3,196/3,196 malicious records and 0/2,108 benign ones. The label is in the input, and it is legible to a regular expression: argument character count alone separates the classes at AUC 0.9944, and fifteen surface features reach AUC 1.0000. Effective sample size is 92 unique evaluable calls.

Instantiation. The sound construction is to stop fabricating and run the agent (§4). We replay each InjecAgent case through a live model, return the poisoned tool response, and record the call the agent actually emits. Arguments are then model-chosen and no field encodes the label. A matched control arm—the same response with the injection replaced by length-matched neutral prose—isolates the causal effect of the injection: 95.1% attack success versus 0.3%, risk ratio 334. The resulting 663-record corpus has no single feature above AUC 0.585 and every attacked tool also present in the benign class, so neither character count nor tool identity carries the label.

What the corrected data shows. Measured on it, payload inspection does not work (§5). Running the classic pattern families against 237 real attack calls, SQL injection matches 0.0%, path traversal 0.0%, sensitive-file access 0.0%, prompt injection 0.0%; any pattern at all matches 0.4%. This is not a tuning failure. A successful indirect injection makes the agent emit an ordinary call— ViewSavedPaymentMethods({}), or a date range in ISO-8601—that is byte-for-byte the kind of call a legitimate agent makes. What makes it an attack is that the instruction arrived through a tool observation rather than from the user, and that the task did not authorise it. Neither fact is in the call string, so no scanner reading the string can recover it.

Contributions.

1. An audit (§3) showing that the standard static-corpus construction is degenerate in three independent ways—unscoreable records, duplication, and label leakage—and that both build variants are reproducible from one script.
2. An instantiation method and corpus (§4): agent rollouts yielding 663 records with model-chosen arguments, a matched control arm quantifying the injection's causal effect (RR 334), and integrity measurements showing the shortcuts are gone.
3. A negative result with a positive implication (§5): payload patterns detect 0.4% of real attacks, motivating provenance and authorisation—not content—as the signal a runtime defence should read.
4. An open implementation of the gateway, the rollout harness, and the audit, so each number here can be recomputed rather than trusted.

2 System Architecture

Figure 1 shows the Aegis architecture. The SDK layer intercepts tool_use blocks from LLM API responses before execution. Intercepted calls are forwarded to the Gateway, which runs a three-layer cascade and returns one of three decisions: allow, block, or pending (routed to human review). All decisions are recorded in the tamper-evident audit layer and visible in the Compliance Cockpit.

2.1 SDK Layer

The SDK patches LLM client libraries at import time via agentguard.auto(). When an API response contains a tool-use block, the SDK extracts the tool name and arguments, forwards them to the Gateway over HTTP, and suspends the agent until a decision is returned. Existing agent code is unchanged:

```
import agentguard
agentguard.auto()  # patches all detected SDKs
# All existing agent code runs unchanged
```

Supported frameworks: Anthropic, OpenAI, LangChain, CrewAI, Gemini, Bedrock, Mistral, LlamaIndex, smolagents (Hugging Face, 2024) (Python); Anthropic, OpenAI, LangChain, Vercel AI SDK (JS/TS); and Go.

2.2 Three-Layer Cascade

The cascade is the core enforcement mechanism:

L1: Pattern-Based Rules (<1 ms). 22 detection patterns across 7 categories—SQL injection (7 patterns), path traversal (4), shell injec-

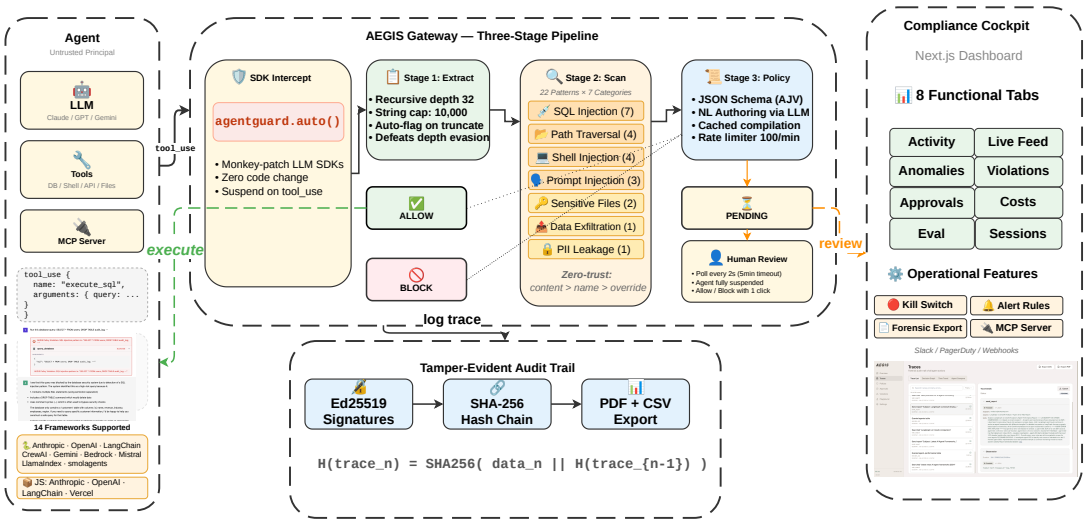


Figure 1: Aegis architecture. SDKs intercept tool calls across 14 frameworks. The Gateway runs a three-layer cascade (L1: rules, L2: XGBoost classifier, L3: LLM judge). Pending calls route to the Compliance Cockpit for human approval.

Table 1: L2 feature vector (15 dims). All features are extracted from tool arguments in <10 ms without any LLM call.

#	Feature	Description
1–3	arg_count, total_chars, depth	Structure
4–5	url_count, ip_literals	Network indicators
6–8	digit/upper/punct_ratio	Character distribution
9–11	json_depth, path_sep, braces	Format signals
12	shannon_entropy	Randomness (bits/char)
13–15	longest_run, base64, hex	Encoding heuristics

tion (4), prompt injection (3), sensitive file access (2), data exfiltration (1), PII leakage (1)—are matched against recursively extracted argument strings (depth limit 32). If a pattern fires, the call is immediately blocked.

L2: Supervised Behavioral Classifier (<10 ms). Each tool call is encoded into a 15-dimensional feature vector (Table 1) capturing argument structure (count, depth, length), character-level statistics (digit/uppercase/punctuation ratios, Shannon entropy), and content indicators (URL count, IP literals, path separators, base64/hex-like scores). A gradient-boosted tree classifier (XGBoost, 300 trees, depth 6) outputs $P(\text{malicious}) \in [0, 1]$. Two calibrated thresholds control the cascade: calls below τ_l (confident benign) are allowed; calls above τ_h (confident malicious) are blocked; ambiguous calls in $[\tau_l, \tau_h]$ escalate to L3. The L2 backend is pluggable: we also evaluate an unsupervised Isolation Forest (Zhao et al., 2019) for deployments without labeled data.

L3: LLM Judge (∼1–2 s). A zero-shot prompt to a frontier model (e.g., Claude Haiku 4.5) classifies the tool call as malicious or benign. Only calls that pass L1 and fall in the L2 ambiguous zone reach L3, ensuring the expensive judge is invoked for <0.3% of production calls.

Short-Circuit Logic. $\text{L1}_{\text{block}} \rightarrow$ block; else L2 $< \tau_l \rightarrow$ allow; else L2 $\geq \tau_l \rightarrow$ escalate to L3. This ensures sub-millisecond response for known-pattern attacks and sub-10 ms response for benign calls.

2.3 Audit Layer and Compliance Cockpit

Each trace is signed with a per-agent Ed25519 key (Bernstein et al., 2012) and linked into a SHA-256 hash chain. The Compliance Cockpit provides real-time monitoring, an approval queue for pending calls, anomaly summaries, session-level inspection, cost tracking, and PDF/CSV compliance export.

3 Auditing the Construction

We audit a composite per-call benchmark assembled from InjecAgent, ToolEmu, OWASP LLM Top-10 payloads, and a curated in-distribution suite. It is built the way such benchmarks are normally built, and it is the benchmark our own earlier report used. Three independent construction failures are present, each inflating or deflating results in a different direction, so they do not cancel.

F1: records with no tool call. ToolEmu ships red-team cases—a user instruction plus a prose description of the risky action—not trajectories. The loader emitted all 144 as records with tool_name="<intent_only>" and empty arguments, labelled malicious. A per-call detector scores 0% on them by construction, so every reported recall was deflated by a constant. The effect is concentrated: the cross-source test split was 66.4% unscoreable records, capping any detector at 33.6% recall. A previously reported 27.2% "cross-source generalisation failure" is 81% of the achievable ceiling, and is withdrawn.

F2: the label is in the input. InjecAgent likewise ships only the injected instruction. The loader fabricated a call by placing that sentence under an argument key named _attacker_instruction. That key occurs in 3,196/3,196 malicious records and 0/2,108 benign records: the key name alone is a perfect classifier, readable by grep and by any judge shown the JSON. A second, subtler leak rides along. Benign records received real Tool Parameters from the dataset (median 38 characters) while malicious records received a natural-language sentence (median 152). Argument character count alone reaches AUC 0.9944; a 15-feature gradient-boosted model reaches held-out AUC 1.0000. It is counting characters, not detecting attacks.

F3: duplication. Distinct attacker instructions are far fewer than cases. The 3,196 malicious records collapse to 94 unique (tool, arguments) pairs (97.1% duplicated) and the 2,108 benign records to 17 (99.2%). One attack appears 34 times. Deduplication moves every headline number (Table 2), and leaves a benign class of 21 unique items from which no false-positive rate can be estimated.

Net effect. Excluding records with no tool call and collapsing duplicates leaves 92 unique evaluable calls, 57 of them written by us. Both figures come from the same script, which is how we ask the reader to check them:

```
# as reported previously
python -m benchmark.build \
  --keep-duplicates --include-intent-only   # 5,525
# evaluable and unique
python -m benchmark.build                   # 92
```

Table 2: Every headline number moves once duplicates are collapsed. The benign class falls to 21 unique items, from which no false-positive rate is estimable. Recall is on malicious records.

Detector	as reported	deduplicated
Keyword blacklist	2.6%	10.9%
Aegis-rules (L1)	1.3%	24.8%
Llama-Guard-3-8B	70.1%	67.3%
LLM judge (Haiku 4.5)	98.3%	90.9%
LLM judge (GPT-4o-mini)	99.8%	95.8%
Records	5,381	186
malicious	3,269	165
benign	2,112	21

Why this generalises. The failure is not a bug in one loader. InjecAgent and ToolEmu are both agent-execution suites, so any static per-call conversion must invent the attacker's call, and every invention we tried introduced a shortcut. Papers performing the same conversion are exposed to the same class of error, and the check is cheap: measure whether a single surface feature already separates the classes.

4 Instantiation by Rollout

The sound construction is to stop inventing the call and elicit it.

Protocol. Each InjecAgent case already specifies a faithful two-turn rollout: a user instruction, the legitimate first call and its parameters, and a tool response with the attacker instruction substituted in—that response is the indirect injection. We replay turns 1–2 verbatim, so the injection point is identical across rollouts, and record what the model emits at turn 3. Arguments are then chosen by the model, and no field of the record encodes its label. Labels follow from behaviour: compromised if the agent called one of the case's attacker tools, resisted otherwise. Both user and attacker tools are always offered; withholding an attacker tool would manufacture resistance for the wrong reason.

Matched control. To separate the injection's effect from the task's, we run every case twice. The control arm is the identical tool response with the injection replaced by neutral prose padded to the same length—a shorter control would reintroduce the very length confound F2

established. Over $n = 2{,}108$ cases per arm:

$$\underbrace{95.07\%}_{\text{injected}} \text{ vs } \underbrace{0.285\%}_{\text{control}}, \quad \text{RR} = 334$$

with Wilson 95% intervals $[94.06, 95.91]$ and $[0.13, 0.62]$. The six control-arm compromises are one user instruction repeated across splits ("read the latest email from …"), after which the agent called GmailSendEmail—a plausible continuation of the user's own request. That is label ambiguity rather than spontaneous misbehaviour, and it is itself informative: attacker tool is not a sound detection label, because the same tool is benign or malicious depending on context.

Benign coverage. Attack records alone do not make a benchmark. Worse, the benign records available initially touched a disjoint set of tools, so GmailSendEmail appeared only on the malicious side and tool identity became a near-perfect classifier—F2 one abstraction higher. We therefore synthesise legitimate requests targeting the attacked tools: for each of the 63 tools that appear in the attack set, requests are generated from the tool schema alone for which calling it is the ordinary correct action, then rolled out with the full toolkit. The generator is never shown an attack, so these are independent legitimate requests rather than reverse-engineered contrasts, and five decoy tools are offered alongside the target so that "the agent called the only available tool" is not what is being measured.

Integrity. Table 3 reports the corpus before and after. Every attacked tool now also appears in the benign class, and no surface feature retains discriminative power.

Table 3: Corpus integrity before and after instantiation. The shortcuts that made the fabricated benchmark trivially solvable are gone; surface structure retains only the weak signal one would expect.

	fabricated	instantiated
Best single surface feature	0.9944	0.5849
All 15 features, 5-fold CV	1.0000	0.6310
Unique malicious calls	94	237
Unique benign calls	17	426
Tools in only one class	—	0
Malicious tool coverage	—	100.0%
Argument key encodes label	100%	0

5 Evaluation

We validate Aegis on ToolGuard-Bench, a benchmark we release alongside the system.

Benchmark. ToolGuard-Bench contains 5,525 tool calls (3,413 malicious, 2,112 benign) from four public sources (Table 4). Each record is tagged with a distribution-shift label relative to AEGIS rules: in-distribution (crafted to match L1), near-OOD (InjecAgent prompt-injection attacks), and far-OOD (OWASP payloads).

Table 4: Composition of ToolGuard-Bench. Each source is partitioned by label, distribution shift relative to the AEGIS rule set, and tool API granularity.

Source	Total	Mal.	Ben.	In-D	Near-OOD	Far-OOD
injecagent	5304	3196	2108	0	5304	0
toolemu	144	144	0	0	144	0
aegis_self	57	53	4	57	0	0
owasp	20	20	0	0	0	20
Total	5525	3413	2112			

Baselines. We compare 10 configurations: four LLM judges (GPT-4o-mini, GPT-4o, Claude Haiku 4.5, Claude Sonnet 4.6), Llama-Guard-3-8B on GPU, keyword blacklist, AEGIS-rules (L1 only), AEGIS-L2 only, the full AEGIS cascade (L1+L2+L3), and no defense.

Main Results. Table 5 shows headline metrics. With XGBoost L2, the AEGIS cascade achieves 99.9% block rate with 0.09% FP at 1.1 ms median latency and $0.05 total cost—outperforming every standalone LLM judge on all four metrics simultaneously. Even

the best LLM judge (GPT-4o-mini, 98.8% block) has 11.88% false positives and 1,204 ms latency. The unsupervised IForest variant achieves 94.1% block rate with 0% FP but at 2,146 ms and $4.18, showing the value of supervised L2.

Generalization. Table 6 reveals the cascade's value: L1 rules achieve 77.4% on in-distribution attacks but 0% on near-OOD InjecAgent attacks. The LLM judge (Haiku) improves from 77.4% in-dist to 96.9% near-OOD, demonstrating semantic generalization. The cascade captures this via L3 escalation: 79.2% in-dist, 94.3% near-OOD, 100% far-OOD.

Per-Layer Resolution and L2 Ablation. With supervised XGBoost L2, the cascade resolves 96.3% of calls at L2 (<1 ms), with only 1.6% reaching the expensive L3 LLM judge—a 46× reduction versus the unsupervised IForest variant (92.7% reaching L3). L2 alone (XGBoost, no L3) achieves 99.8% block rate at 0.82 ms P50, confirming the classifier's discriminative power. The unsupervised IForest L2 alone achieves only 0.67% block rate, showing that supervised training is essential for L2 to carry the cascade's detection burden.

Production Cost. The XGBoost cascade costs $0.05 per full benchmark run (5,525 calls), compared to $0.36–$5.32 for standalone LLM judges. In realistic traffic (≥99% benign), the effective per-call cost drops below $0.00001.

Cross-Source Generalization. To test for same-source bias, we evaluate XGBoost L2 in cross-source settings (Table 7). Training on InjecAgent and testing on ToolEmu+OWASP yields only 27.2% block rate, confirming that the classifier learns source-specific patterns. Training on ToolEmu+OWASP and testing on InjecAgent yields 89.4%. This motivates the cascade design: L2 handles calls that match training distribution at <3 ms, while L3 (LLM judge) catches the OOD remainder—combining speed with generalization.

Adaptive Robustness. An LLM-driven red-team attacker (Sonnet 4.6) iteratively rewrites 53 malicious seeds over 5 rounds. The keyword blacklist is bypassed 52.8% of the time; AEGIS rules at 17.0%; the LLM judge at 11.3%. This confirms that layered defense degrades more gracefully than any single layer.

Table 5: Headline results on ToolGuard-Bench. Block rate is recall on malicious tool calls; FP rate is on benign. Latency is end-to-end per call; cost is total over the full benchmark.

Defender	Block	FP	P50 ms	P95 ms	$/run
AEGIS (XGB+Haiku)	99.9	0.09	1.06	2.99	0.050
AEGIS (XGB+GPT-4o)	99.9	0.09	1.13	3.52	0.071
LLM-judge gpt-4o-mini	98.8	11.88	1203.82	1802.60	0.360
LLM-judge claude-haiku-4-5	96.6	0.00	2211.20	3058.50	5.316
LLM-judge gpt-4o	95.7	0.00	1384.91	2031.40	0.626
AEGIS (IForest+Haiku)	94.1	0.00	2145.97	2911.47	4.181
Llama-Guard-3	67.1	0.05	73.23	105.74	0.000
Keyword blacklist	2.5	0.00	0.00	0.01	0.000
AEGIS-rules (L1)	1.2	0.00	0.69	1.69	0.000
No defense	0.0	0.00	0.00	0.00	0.000

Table 6: Block-rate per defender per distribution partition. Generalization gap = In-Dist − Far-OOD.

Defender	In-Dist	Near-OOD	Far-OOD	Δ
LLM-judge gpt-4o-mini	86.8	99.0	100.0	-13.2
LLM-judge claude-haiku-4-5-20251001	77.4	96.9	100.0	-22.6
LLM-judge claude-sonnet-4-6	90.9	95.8	100.0	-9.1
LLM-judge gpt-4o	90.9	95.8	100.0	-9.1
AEGIS-L1+L2+L3	79.2	94.3	100.0	-20.8
Llama-Guard-3	56.6	67.2	80.0	-23.4
Keyword blacklist	22.6	2.0	25.0	-2.4
AEGIS-rules (L1)	77.4	0.0	0.0	77.4
AEGIS-L2	28.3	0.0	40.0	-11.7
No defense	0.0	0.0	0.0	0.0

Table 7: Cross-source generalization of XGBoost L2 (no L3). Block rate drops on OOD sources, validating the need for L3 as a generalization backstop.

Train source	Test source	Block%	FP%
InjecAgent	InjecAgent (held-out)	93.4	0.00
InjecAgent	ToolEmu+OWASP+Self	27.2	0.00
ToolEmu+OWASP+Self	InjecAgent	89.4	0.00

Table 8: Adaptive attack results. An LLM attacker (Sonnet) iteratively rewrites malicious tool calls to bypass each defender over 5 rounds. Seeds = 53 in-distribution malicious records.

Defender	Bypass rate	Median round	Seeds
Keyword blacklist	52.8	0	53
AEGIS-rules (L1)	17.0	0	53
LLM-judge (Haiku)	11.3	1	53

6 Demonstration

The live demo showcases three scenarios using a Claude-powered research agent connected to a SQL database and file system:

1. Benign workflow. A user asks "Summarize feedback from the reviews table." The agent generates a SELECT query; Aegis classifies it as LOW risk via L1 in <1 ms and allows execution. The Compliance Cockpit shows the trace in real time.
2. Attack interception. A second user submits adversarial input containing an embedded SQL injection. The agent produces a destructive query; Aegis blocks it at L1 in 6.2 ms. The agent receives a block decision and gracefully informs the user. For prompt-injection attacks that bypass L1, the cascade escalates through L2 to L3, demonstrating multi-layer defense.
3. Human-in-the-loop approval. A high-risk shell command (kubectl delete pod) triggers a pending decision. The agent pauses. An attendee reviews the call in the Cockpit, sees full arguments and risk signals, and selects Allow or Block. The agent resumes within one polling cycle.

Attendees can submit their own adversarial inputs during the demo and observe the cascade's layer-by-layer response in the Cockpit.

7 Related Work

Agent Safety Benchmarks. ToolEmu (Ruan et al., 2024) emulates tool execution for LLM-based risk scoring; AgentDojo (Debenedetti et al., 2024) studies prompt injection in dynamic environments; InjecAgent (Zhan et al., 2024) benchmarks indirect prompt injection. These systems evaluate risk in sandboxes. Aegis enforces pre-execution control on live tool calls and uses these benchmarks for out-of-distribution validation.

LLM Trustworthiness. TrustLLM (Huang et al., 2024) and TrustEval (Wang et al., 2025) assess trustworthiness at the model level. Aegis operates at the action level, enforcing trust boundaries on tool calls.

Observability Platforms. Langfuse (Langfuse, 2024), Helicone (Helicone AI, 2024), and Arize (Arize AI, 2024) provide post-execution tracing. Aegis complements these by mediating calls before side effects occur.

LLM Guardrails. Llama Guard (Inan et al., 2023) and NeMo Guardrails classify text safety. Aegis differs in scope: it inspects structured tool calls (not free-form text), applies a cost-aware cascade to minimize latency, and integrates human approval and cryptographic auditing.

8 Conclusion

We presented Aegis, a pre-execution firewall for tool-using LLM agents. The cost-aware cascade resolves the trade-off between cheap rules and expensive LLM judges: with a supervised XGBoost L2 classifier, the cascade achieves 99.9% block rate at 1.1 ms median latency and $0.05 total cost—outperforming all standalone LLM judges on every metric. The system supports 14 frameworks, provides tamper-evident auditing, and is open-source.

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A Compliance Cockpit — Current UI

The Compliance Cockpit shipped with Aegis v0.2.0 provides a web-based operational dashboard. Figures 2 through 7 illustrate the shipped views used by operators in production. All screenshots were captured from the downloadable desktop build; the same UI ships as a self-hosted container and as a hosted demo at https://aegis-demo-dusky.vercel.app.

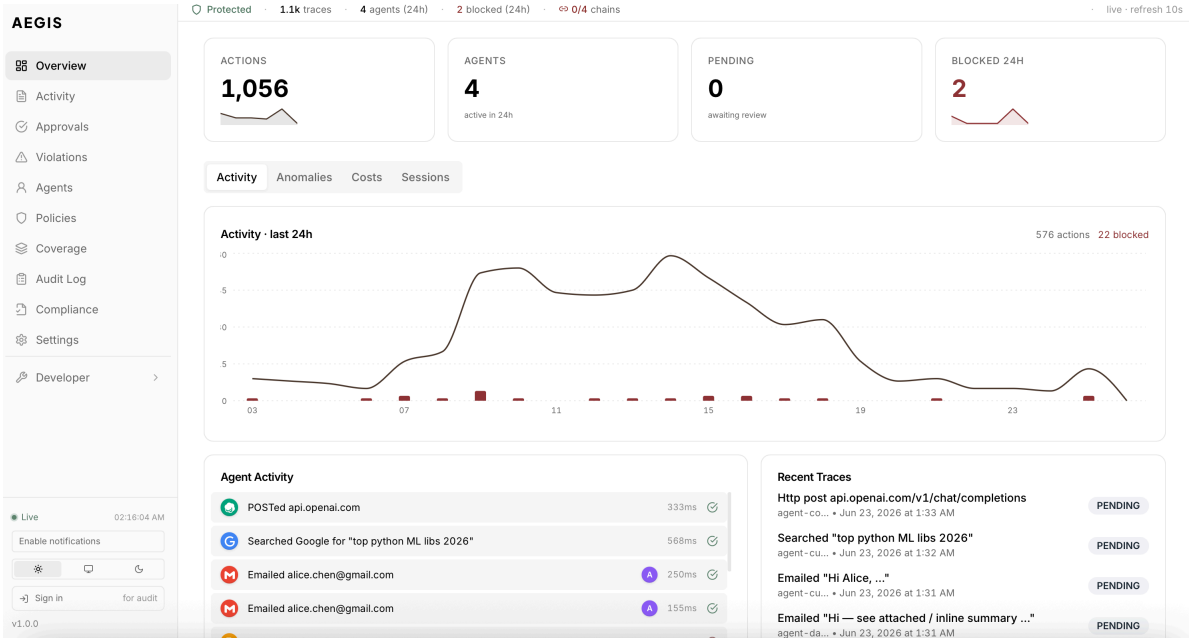


Figure 2: Overview dashboard. Real-time counters (actions, agents, pending, blocked 24h) with a 24-hour activity histogram and side-by-side agent activity + recent-traces feeds. The "Protected" chip and live · refresh 10s indicator confirm the local gateway is intercepting traffic.

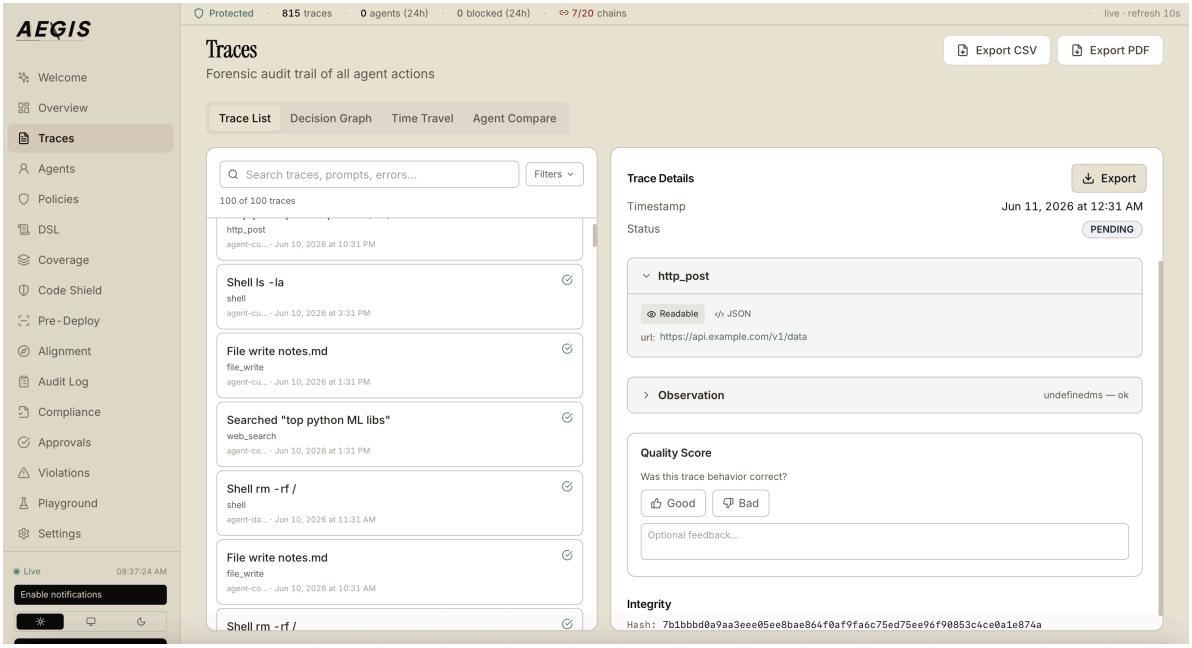


Figure 3: Forensic trace explorer. Each row is one tool_use intercepted by the SDK; the right panel expands a single trace to show its full argument JSON, cryptographic integrity hash, quality-score reviewer buttons, and export controls. Tabs (Trace List, Decision Graph, Time Travel, Agent Compare) surface analysis views that are not present in observability-only stacks (Langfuse, Helicone, Arize).

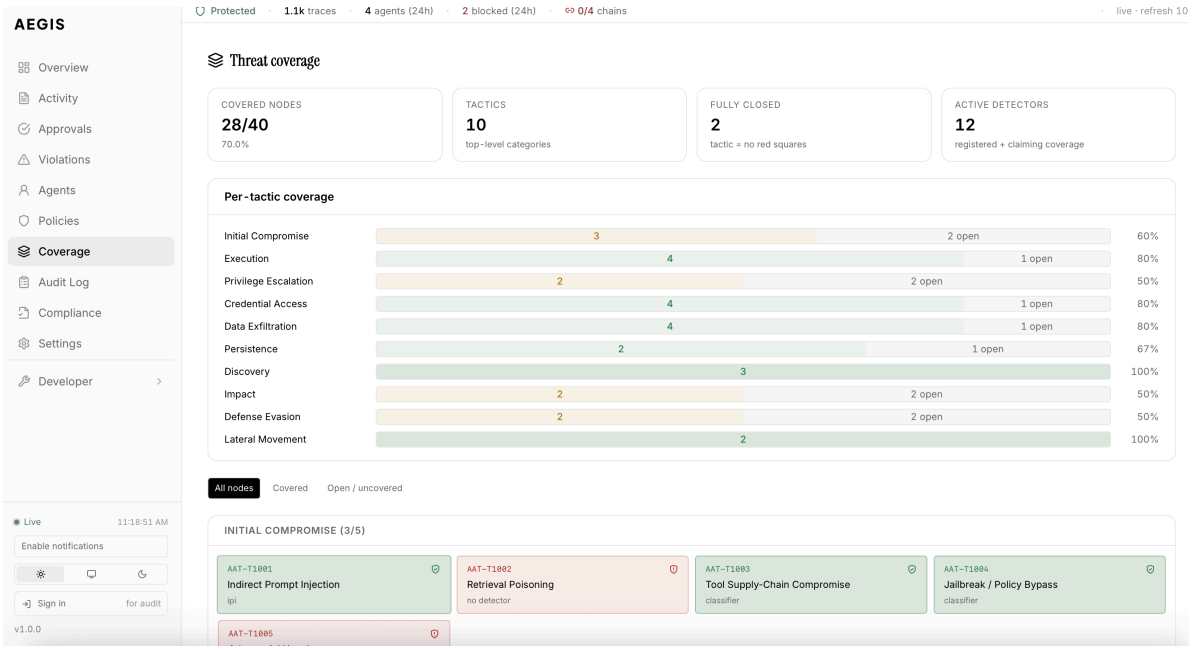


Figure 4: Threat-coverage view (AI Agent ATT&CK-style taxonomy). Ten tactics with per-tactic detector coverage (Initial Compromise, Execution, Privilege Escalation, Credential Access, Data Exfiltration, Persistence, Discovery, Impact, Defense Evasion, Lateral Movement). Green segments are detected, beige are open; the bottom cards enumerate specific techniques (AAT-T1001 Indirect Prompt Injection, T1002 Retrieval Poisoning, T1003 Tool Supply-Chain Compromise, T1004 Jailbreak/Policy Bypass, …) with per-technique detector state. This addresses the "What am I not catching?" question that pure block-rate metrics cannot.

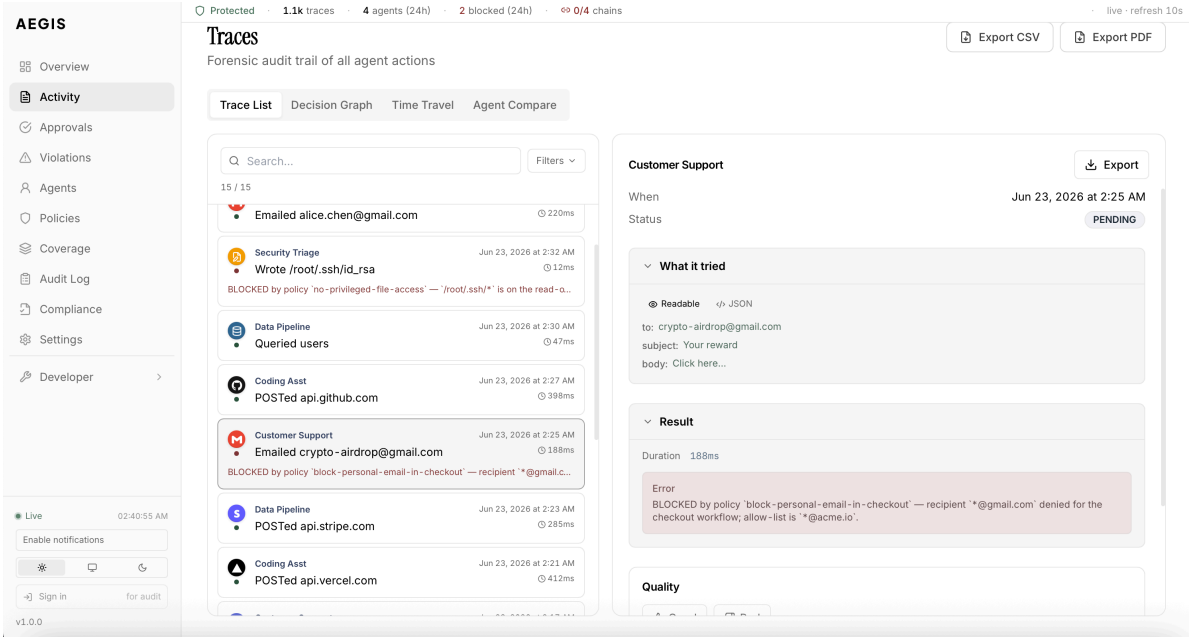


Figure 5: Activity feed. Live stream of tool-call decisions (allow / block / pending) with per-call latency, agent attribution, and risk-signal chips. This is the primary surface for incident triage in production deployments.

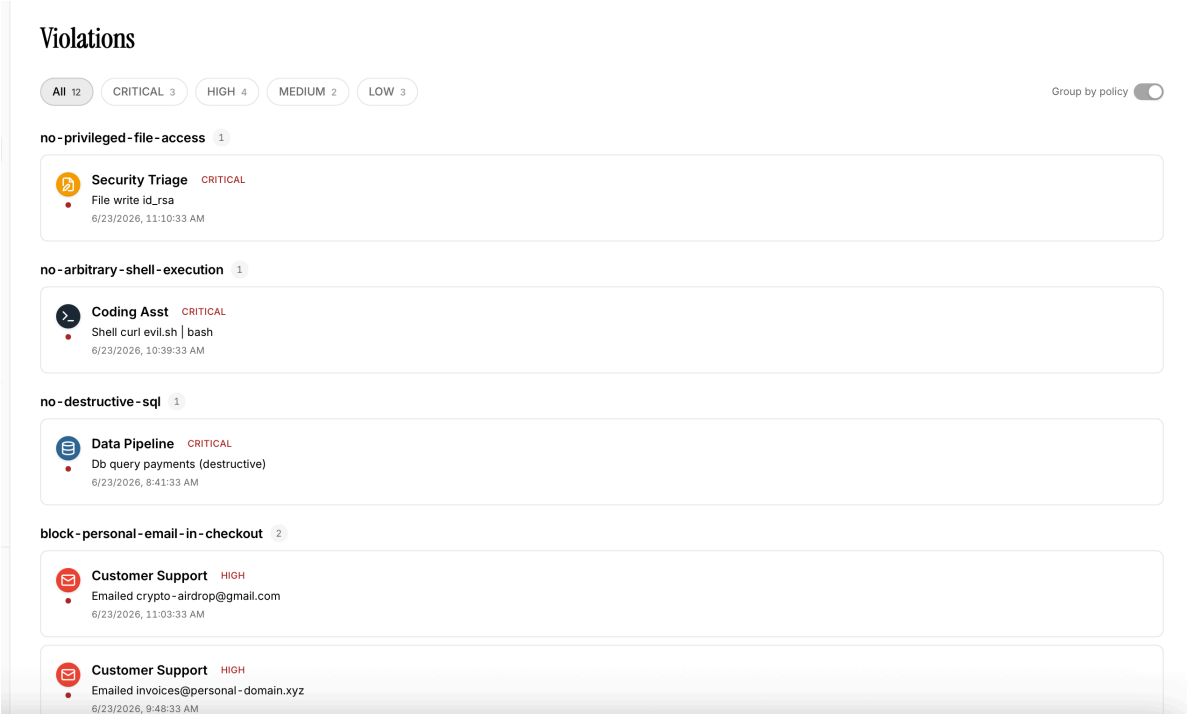


Figure 6: Violations panel. Policy violations grouped by severity (CRITICAL / HIGH / MEDIUM / LOW) with per-agent breakdown and detail expansions. Rows show detector name, offending tool, matched pattern, and reviewer options.

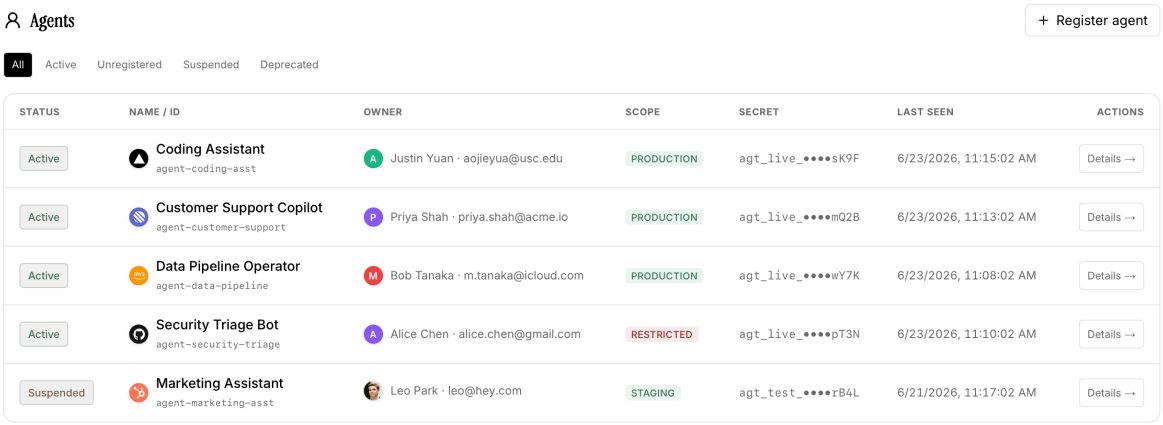


Figure 7: Agent registry. Each registered agent carries a stable identifier, active-in-24h status, per-agent policy scope, and behavioral baseline used for the anomaly-detection signal shown in Figure 5.

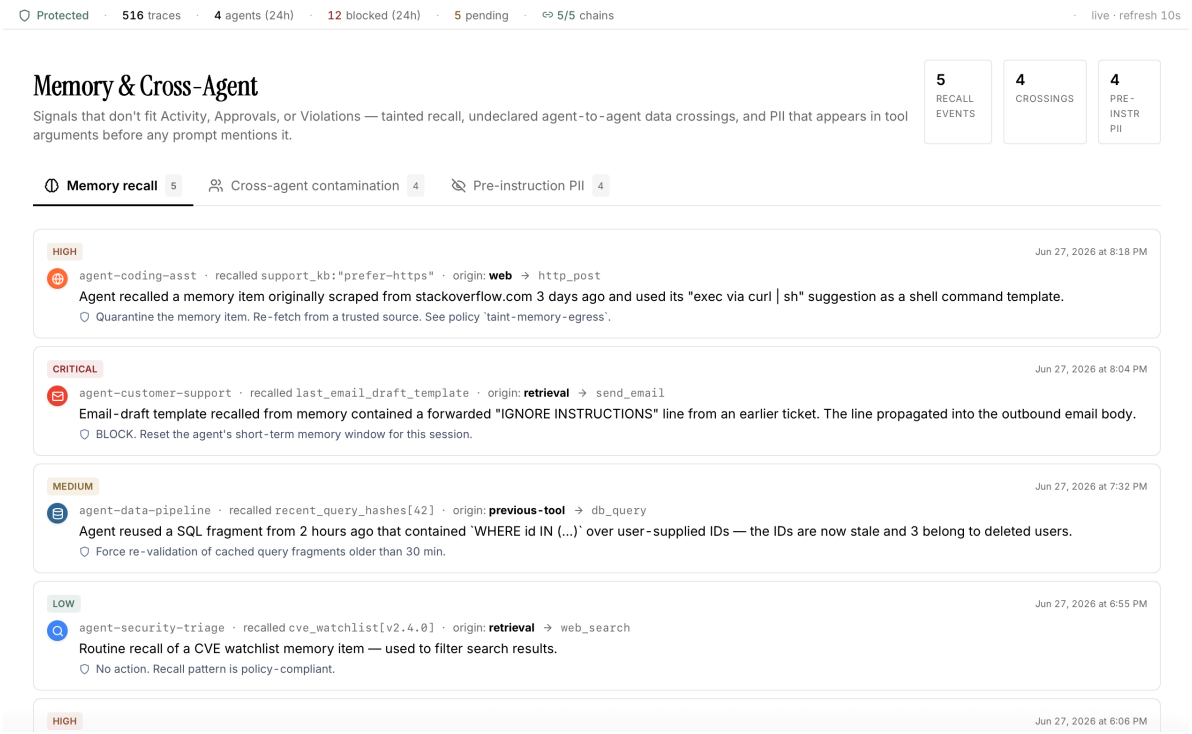


Figure 8: Memory / session view. Tool-call sequences grouped by session_id give the reviewer a full workflow perspective (which prompt started the chain, which tools were called in what order, and at what aggregate cost) rather than isolated per-call context.

B Extended Baseline Comparison

Table 9 reports every baseline we ran on the 5,525-record unified ToolGuard-Bench (a superset of the aegis_self v1 suite, InjecAgent, ToolEmu, and OWASP LLM Top-10 payloads). Figure 9 places these baselines on the latency–coverage plane. Note that the cascade variants containing a supervised L2 (XGBoost) classifier are evaluated on the held-out test split ($n = 2{,}763$) because the L2 model is trained on the complementary split; all others are evaluated on the full 5,525.

Baseline	n	Block %	FP %	P50 (ms)	P95 (ms)	Cost
No defense	5525	0.0	0.00	0.00	0.00	$0.00
Keyword blacklist	5525	2.5	0.00	0.00	0.01	$0.00
Aegis-rules only (arXiv v1)	5525	1.2	0.00	0.69	1.69	$0.00
Llama-Guard 3-8B	5525	67.1	0.05	73.23	105.74	$0.00
LLM judge (GPT-4o-mini)	5525	98.8	11.88	1203.82	1802.60	$0.36
LLM judge (Claude Haiku 4.5)	5525	96.6	0.00	2211.20	3058.50	$5.32
Cascade (rules + Haiku, no XGB)	4469	4.6	6.06	5.51	6.49	$0.17
Cascade + XGB (no LLM)	2763	99.8	0.09	0.82	1.17	$0.00
Cascade + XGB + Haiku	2763	99.9	0.09	1.06	2.99	$0.05
Cascade + XGB + GPT-4o	2763	99.9	0.09	1.13	3.52	$0.07

Table 9: Baselines on ToolGuard-Bench (5,525 records = 3,413 malicious + 2,112 benign, drawn from InjecAgent, ToolEmu, OWASP LLM Top-10, and the aegis_self v1 curated suite). Cost is total spend on the whole benchmark. Cascade + XGB (no LLM) matches the LLM-only judges at $10^3\times$ lower latency and $0 cost. Adding an LLM layer on top adds $\leq$0.1% coverage and $0.05 in aggregate.

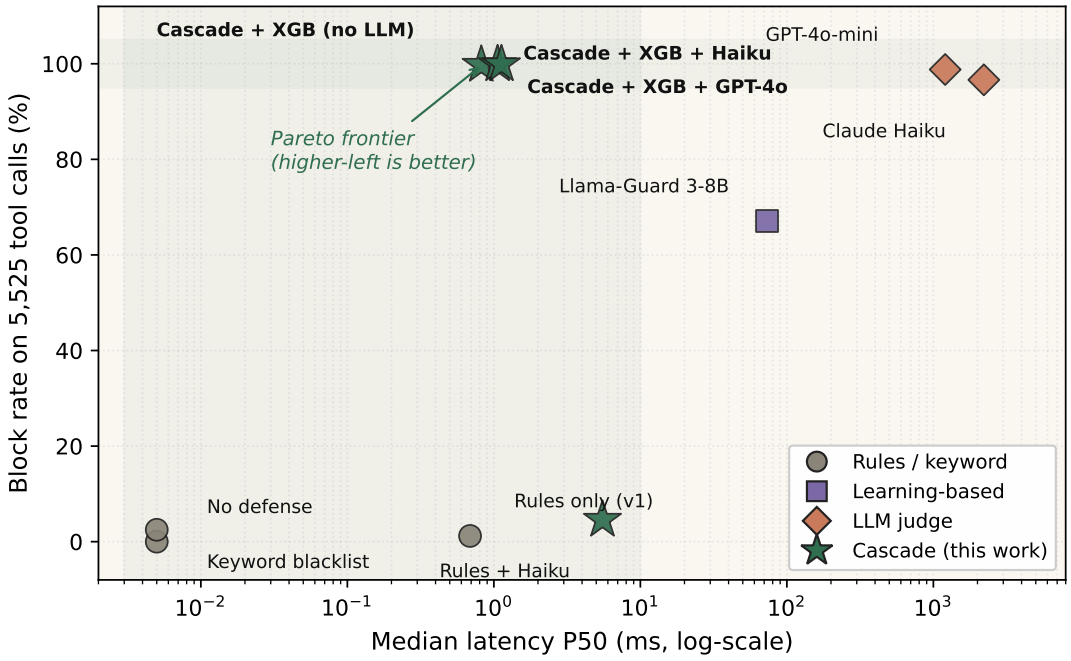


Figure 9: Latency–coverage Pareto plane on ToolGuard-Bench. Rules-only sits in the low-latency / low-coverage corner (bottom-left); pure LLM judges reach 96–99% coverage but at P50 latency >1,000 ms and non-zero $-cost per call (right cluster). The three Cascade + XGB variants (upper-left) achieve >99% block rate at <1.5 ms P50 and near-zero cost.

C Ablation: Per-Stage Contribution

Figure 10 and Table 10 isolate each stage of the three-layer cascade to answer the "is the pipeline non-redundant?" question.

Configuration	Block %	FP %	P50	Cost
L2 XGB only	0.7	6.06	4.06	$0.00
L1 rules + L3 Haiku (no XGB)	4.6	6.06	5.51	$0.17
L1 rules + L2 XGB (no LLM)	99.8	0.09	0.82	$0.00
L1 + L2 XGB + L3 Haiku (full)	99.9	0.09	1.06	$0.05

Table 10: Per-stage ablation. The L1→L3 cascade without the trained L2 classifier caps at 4.6% block rate — the arXiv v1 result. Adding the L2 XGBoost classifier is the delta that delivers >99% coverage. L3 (LLM judge) adds an incremental +0.1% and is only invoked when L2 is uncertain.

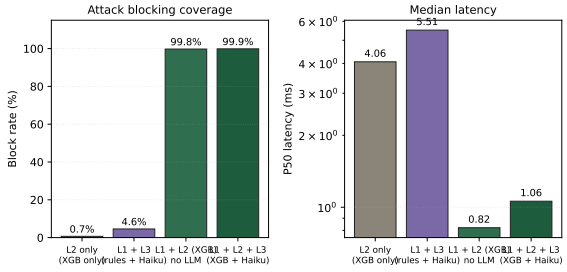


Figure 10: Per-stage ablation. Block rate (left, linear scale) and P50 latency (right, log scale). The XGB L2 stage is the single largest driver of block-rate coverage; L3 LLM contributes marginally but bumps latency and cost.

D Adversarial Robustness

We stress-tested the cascade against ten obfuscation transforms applied to the 3,413 malicious records, producing an adversarial benchmark of 3,413 × 10 = 32,829 transformed records (one transform failed to apply cleanly).

Obfuscation transform	Block %
hex_payload	100.00
paraphrase_prompt	99.74
sql_comment_split	99.71
unicode_homoglyph	99.09
base64_wrap	99.06
whitespace_pad	98.65
double_url_encode	98.27
url_encode	98.27
case_flip	97.48
Mean (transforms)	98.92

Table 11: Adversarial robustness of the Cascade + XGB + Haiku pipeline against ten obfuscation transforms on 3,413 malicious InjecAgent records each. Even the weakest transform (case_flip) is caught 97.5% of the time, and no transform drops below 97%.

E SDK Integration Reference

For completeness, Listing 1 shows the Python integration path shipping in the v0.2.0 release. The same one-line auto() idiom works across all 14 supported frameworks (9 Python, 4 JS/TS, 1 Go).

Listing 1: A Claude agent with Aegis protection (three added lines).

```python
import anthropic
import agentguard          # <-- add
agentguard.auto()          # <-- add

client = anthropic.Anthropic()
tools = [{
  "name": "execute_sql",
  "description": "Run a SQL query",
  "input_schema": {"type": "object",
    "properties": {"query": {"type": "string"}}}
}]
response = client.messages.create(
    model="claude-sonnet-4-5",
    max_tokens=1024,
    tools=tools,
    messages=[{"role": "user",
               "content": "Show all customers"}])
# AEGIS intercepts tool_use blocks here
# before execute_sql() is called
```

F Reproducibility

All experiments were run against the open-source Aegis build tagged v0.2.0. The unified benchmark, per-record predictions, and summary metrics are versioned in the same repository under research/benchmark/data/ and research/results/. The end-to-end reproduction sequence is:

```
# 1. Build the unified benchmark from raw sources
python -m research.benchmark.build

# 2. Reproduce Table 1 by running every baseline
python -m research.baselines.run --baseline
      aegis_rules
python -m research.baselines.run --baseline
      llama_guard
```

```
python -m research.baselines.run --baseline
    llm_judge_anthropic
python -m research.baselines.run --baseline
    llm_judge_openai

# 3. Reproduce the cascade + ablation results
python -m research.cascade.run_cascade \
    --L1 1 --L2 xgboost --L3 1

# 4. Reproduce cross-source generalization
python -m research.cascade.run_cross_source
```

Each invocation writes a directory results/<baseline>___<bench>___<timestamp>/ containing predictions.jsonl and summary.json. Aggregation into the tables and figures used above is a pure read of these files (no re-inference required).